Homework 5: Adaptive Non-stiff ODE Solver

Problem Formulation

In this homework assignment, we aim to develop a solver for non-stiff ordinary differential equations. That is, ODEs of the form

$$\mathbf{y}' = \mathbf{f}(t, \mathbf{y}) \tag{1}$$

that satisfy $h\lambda \gg -\infty$.

I will use second-order Adams linear multistep methods, utilizing P(EC)E to reduce computational cost of implicit Adams-Moulton with Adams-Bashforth as the predictor method. I will also use adaptive step size control to obtain a numerical solution to (1) within a target error tolerance. Finally, I will use an interpolating polynomial to approximate the numerical solution at specific time points.

Numerical Schemes

A Linear Multistep Method is a numerical method which involves a linear combination of multiple steps to approximate the solution at the next time step. For non-constant step size Adams methods, we simply integrate (1) across a single timestep of size h_n

$$\mathbf{y}_n = \mathbf{y}_{n-1} + \int_{t_{n-1}}^{t_n} \mathbf{f}(t, \mathbf{y}(t))\, dt$$

The Adams methods approximate this integral using a polynomial interpolation of $\mathbf{f}(t, \mathbf{y}(t))$ using the past values of $\mathbf{f}$. One option for the interpolating polynomial is the Newton interpolation. The Newton interpolating polynomial passing through a set of points at times t_i with the coefficients computed by the divided difference table, defined recursively by

$$\mathbf{f}[t_n, t_{n-k}] = \frac{\mathbf{f}[t_n, t_{n-k+1}] - \mathbf{f}[t_{n-1}, t_{n-k}]}{t_n - t_{n-k}} \tag{2}$$

where $\mathbf{f}[t_n] = \mathbf{f}(t_n, \mathbf{y}(t_n))$. The polynomial is computed with

$$\phi(t) = \mathbf{f}[t_n] + \mathbf{f}[t_n, t_{n-1}](t - t_n) + \mathbf{f}[t_n, t_{n-2}](t - t_n)(t - t_{n-1}) + \cdots + \mathbf{f}[t_n, t_{n-k}](t - t_n) \cdots (t - t_{n-k+1}) \tag{3}$$

For second order Adams-Bashforth, we interpolate the past two $\mathbf{f}$'s. Integrating, we have the Adams-Bashforth method for non-constant step size

$$\mathbf{y}_n = \mathbf{y}_{n-1} + h_n \mathbf{f}_{n-1} \qquad \text{order 1} \tag{4}$$

$$\mathbf{y}_n = \mathbf{y}_{n-1} + h_n \mathbf{f}_{n-1} + \frac{h_n^2}{2h_{n-1}}(\mathbf{f}_{n-1} - \mathbf{f}_{n-2}) \qquad \text{order 2} \tag{5}$$

For Adams-Moulton, we interpolate across the current $\mathbf{f}$ first, and then the past $\mathbf{f}$'s. Thus, the Adams-Moulton method is given by the implicit scheme

$$\mathbf{y}_n = \mathbf{y}_{n-1} + h_n \mathbf{f}_n \qquad \text{order 1} \tag{6}$$

$$\mathbf{y}_n = \mathbf{y}_{n-1} + \frac{h_n}{2}(\mathbf{f}_n + \mathbf{f}_{n-1}) \qquad \text{order 2} \tag{7}$$

In order to avoid the computational cost and complexity of implementation introduced by the implicit Adams-Moulton method, we instead use the Adams-Bashforth method to predict $\mathbf{y}_n$ that is required by $\mathbf{f}_n$ in the Adams-Moulton method. The Predictor-Evaluate-Correct-Evaluate method first predicts $\mathbf{y}_n \approx \mathbf{y}_n^{(0)}$, which is only a good approximation in the non-stiff limit. $\mathbf{f}(t, \mathbf{y})$ is then evaluated at $(t_n, \mathbf{y}_n^0)$. The Corrector method uses this approximation to determine $\mathbf{y}_n$, after which $\mathbf{f}(\mathbf{t}, \mathbf{y})$ is evaluated at $(t_n, \mathbf{y}_n)$. The second-order Predictor-Corrector algorithm is as follows:

$$\begin{aligned}
&\text{Predict} \quad \mathbf{y}_n^{(0)} = \mathbf{y}_{n-1} + h_n \mathbf{f}_{n-1} + \frac{h_n^2}{2h_{n-1}}(\mathbf{f}_{n-1} - \mathbf{f}_{n-2}) \\
&\text{Evaluate} \quad \mathbf{f}_n^{(0)} = \mathbf{f}(t_n, \mathbf{y}_n^{(0)}) \\
&\text{Correct} \quad \mathbf{y}_n = \mathbf{y}_{n-1} + \frac{h_n}{2}(\mathbf{f}_n^{(0)} + \mathbf{f}_{n-1}) \\
&\text{Evaluate} \quad \mathbf{f}_n = \mathbf{f}(t_n, \mathbf{y}_n)
\end{aligned} \tag{8}$$

However, on the first time step, we only have the point $(t_0, \mathbf{y}(t_0))$, so we cannot use a two-step Adams-Bashforth method. In this case, we determine the starting values by using the one-step Adams-Bashforth method as the predictor and the two-step Adams-Moulton method as the corrector.

$$\begin{aligned}
&\text{Predict} \quad \mathbf{y}_n^{(0)} = \mathbf{y}_{n-1} + h_n \mathbf{f}_{n-1} \\
&\text{Evaluate} \quad \mathbf{f}_n^{(0)} = \mathbf{f}(t_n, \mathbf{y}_n^{(0)}) \\
&\text{Correct} \quad \mathbf{y}_n = \mathbf{y}_{n-1} + \frac{h_n}{2}(\mathbf{f}_n^{(0)} + \mathbf{f}_{n-1}) \\
&\text{Evaluate} \quad \mathbf{f}_n = \mathbf{f}(t_n, \mathbf{y}_n)
\end{aligned} \tag{9}$$

For constant step size, the previous equations are still used, but with the additional fact that $h_0 = h_1 = \cdots = h_n$.

Step Size Control

The step size is accepted if the Local Truncation Error is below the provided error tolerance ETOL. The LTE is estimated by the difference between the predictor and corrector methods

$$\text{LTE} \approx \left| \frac{C_{p+1}}{C_{p+1} - \hat{C}_{p+1}} \right| \|\mathbf{y}_{\text{pred}} - \mathbf{y}_{\text{corr}}\| \tag{10}$$

where C_{p+1} is the factor in the LTE estimation of the method

$$\mathbf{d}_n = |C_{p+1} h^p y^{(p+1)}| + \mathcal{O}(h^{p+1}) \tag{11}$$

and $\hat{C}_{p+1}$ for the predictor method and C_{p+1} is the factor for the corrector method.

For Adams-Bashforth, the predictor, $\hat{C}_{p+1} = -\frac{1}{12}$. For Adams-Moulton, the corrector, $\hat{C}_{p+1} = \frac{5}{12}$. Thus, the LTE estimate is

$$\text{EST} \approx \frac{1}{6}\|\mathbf{y}_{\text{pred}} - \mathbf{y}_{\text{corr}}\|. \tag{12}$$

We reject a step if EST > ETOL. Then we adjust the step size by

$$h_n \leftarrow r \cdot h_n \tag{13}$$

where

$$r^{p+1} = \frac{\text{frac} \cdot \text{ETOL}}{\text{EST}} \tag{14}$$

which results in a decrease in step size because EST is larger. If we accept a step, then EST < ETOL and we use (14) as the initialization for the next step size selection iteration, thus resulting in an increase in step size due to smaller value of EST.

Evaluating the Solution at Specific Times

Examples

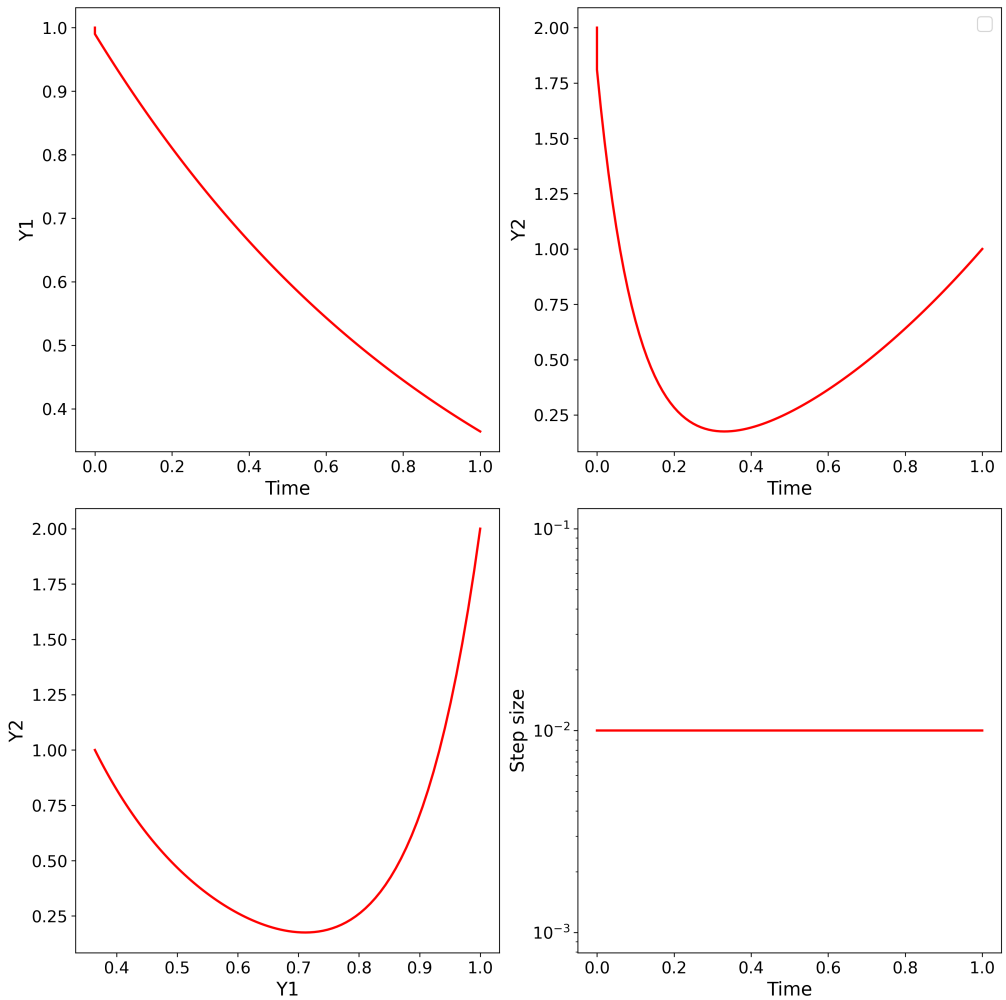

Figure 1: Constant step size

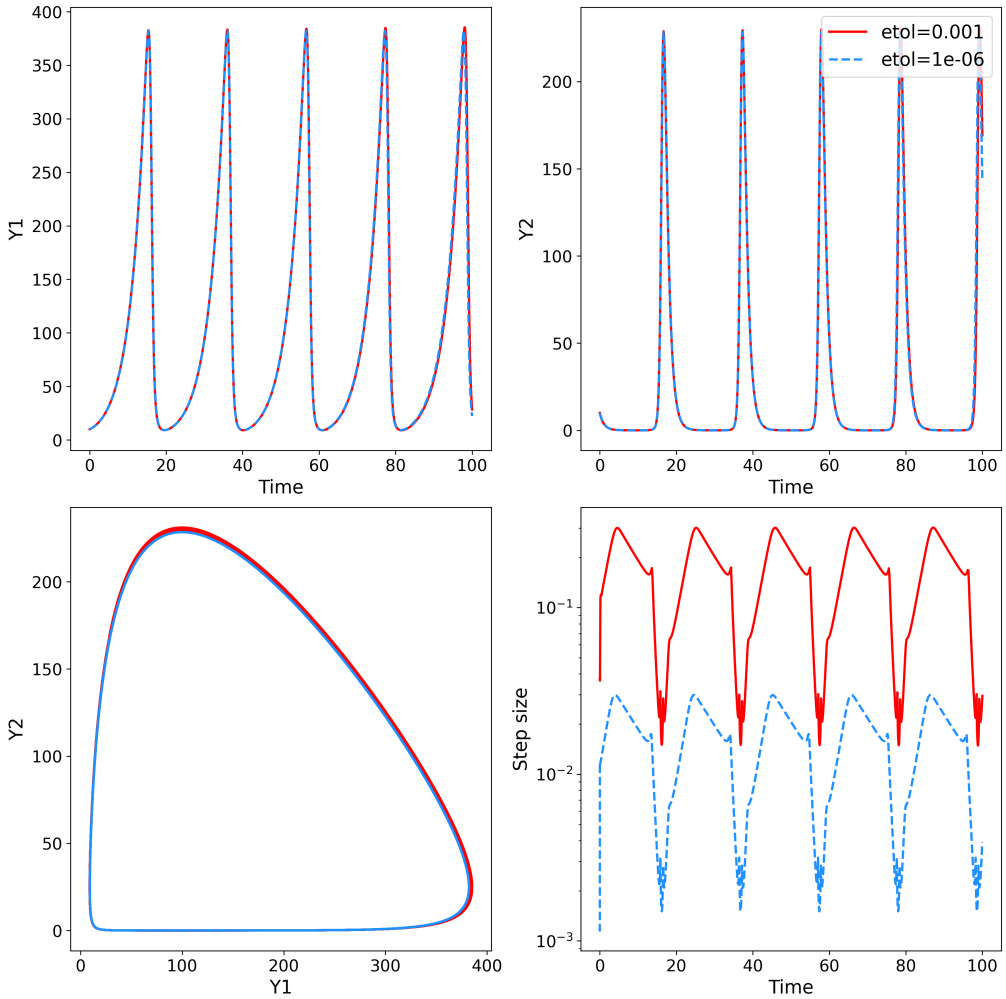

Figure 2: Predator-Prey problem

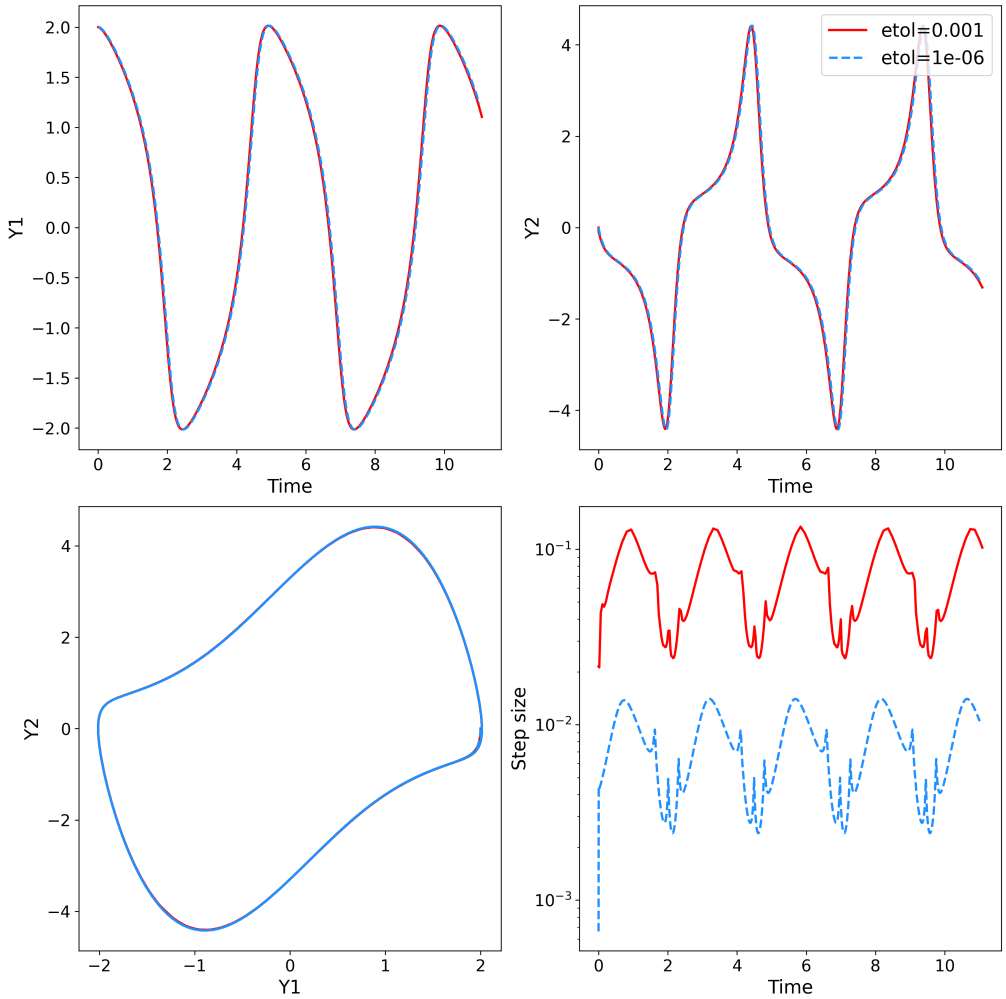

Figure 3: Van der Pol Oscillator

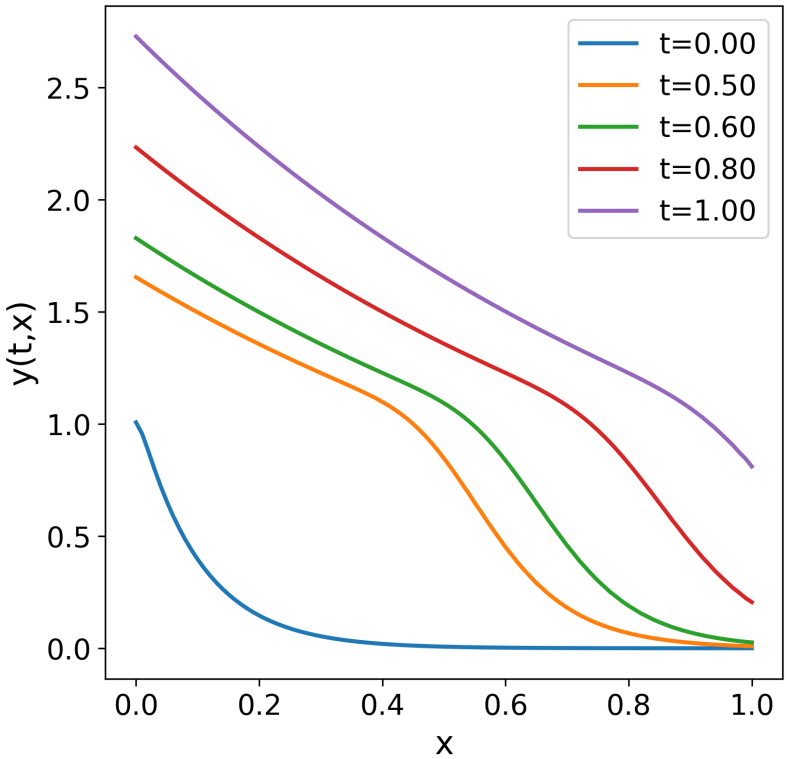

Figure 4: Method of Lines